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RESEARCH PAPER

Near Infrared (NIR) Spectroscopy: The green analytical tool for tea quality analyses – A short review

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ABSTRACT

Tea is a common beverage for a large population worldwide. It is a complex matrix of innumerable chemical compounds which produce a synergistic effect to give the distinctive quality attributes of the tea brew. With the increased public attention in tea quality in recent decades, the demand for high quality teas obviously require high standards of quality assurance and process control. Conventionally, tea quality is assessed by a human panel of tasters to rank and decide its market price. Quality estimations by chemical procedures are precise, but are invasive, skill requiring, time consuming and require costly chemicals and instruments. The application of non-invasive, simple yet fast green electronic techniques has been a growing area of research in recent years. Amongst them, near-infrared (NIR) spectroscopy has gained wide acceptance over conventional analytical and sensory techniques by virtue of its advantages as a fast, reliable and non-destructive analytical tool for qualitative and quantitative measurement of quality attributes.

This paper gives a brief overview of the basic principles, scope, challenges and future of NIR spectroscopy and some recent advances in its application to tea. The paper also provides an insight of the initial work on NIR at Tocklai Tea Research Institute in collaboration with Jadavpur University, Kolkata.

Keywords: Near Infrared Spectroscopy, Diffuse Reflectance, Chemometrics

INTRODUCTION

What is NIR?

The best overview of NIR was made by Williams and Norris:

"Near-infrared reflectance spectroscopy involves optical data, the amount of radiation reflected from a sample relative to the radiation striking it. If you try to look into a room from the outside on a sunny day, it is not easy to see the objects inside the room. Most of the light is reflected back into your eyes, so all that your brain records is the shape of the window. This reflected light is specular reflectance or Fresnel reflectance. If you shade your eyes so that the sunlight is not reflected directly back at you, you will be able to see into the room and your eyes (the detectors) will receive light from the same light source (the sun) reflected back from objects inside the room. This is analogous to diffuse reflectance, since the light reaching your eyes carries information reflected back from the objects inside the room. Your brain (the computer) records what is present in the room because years of observation (calibration) have programmed your brain to recognize familiar signals in terms of colours, shades, furniture, pictures, appliances, people, pets, and other things, even brand names of appliances, names of artists, breeds of dogs, etc."

Near-infrared (NIR) refers to the spectral region that extends from 780 nm to 2,500 nm (wavenumber

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